eSim Semester Long Internship – Autumn 2026

Task 7: PCB Design Using eSim

SMART SOLAR BATTERY CHARGING, PROTECTION & MONITORING PCB

Software Used: eSim Version 2.5
PCB Type: Two-Layer Printed Circuit Board
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Github repo:
https://github.com/techier-coder-4/Smart-Solar-Battery-Charging-Protection-Monitoring-System

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1. Project Overview

The Smart Solar Battery Charging, Protection & Monitoring PCB is a two-layer PCB developed using eSim EDA environment. The project combines power-input handling, protection, battery-related circuitry, voltage regulation and visual status indication into a single PCB design.

2. Problem Statement

Small standalone electronic systems (sensor nodes, garden lighting, portable devices) often need to run on solar power with a rechargeable battery, but a raw solar panel cannot be connected directly to a battery without risking overcharging, reverse-polarity damage, or an unregulated, unsafe output. This project addresses that by combining input protection, correct single-cell battery charging, voltage monitoring, status indication, and output regulation on one small board.

3. Problem Objectives

The objectives of the project are:

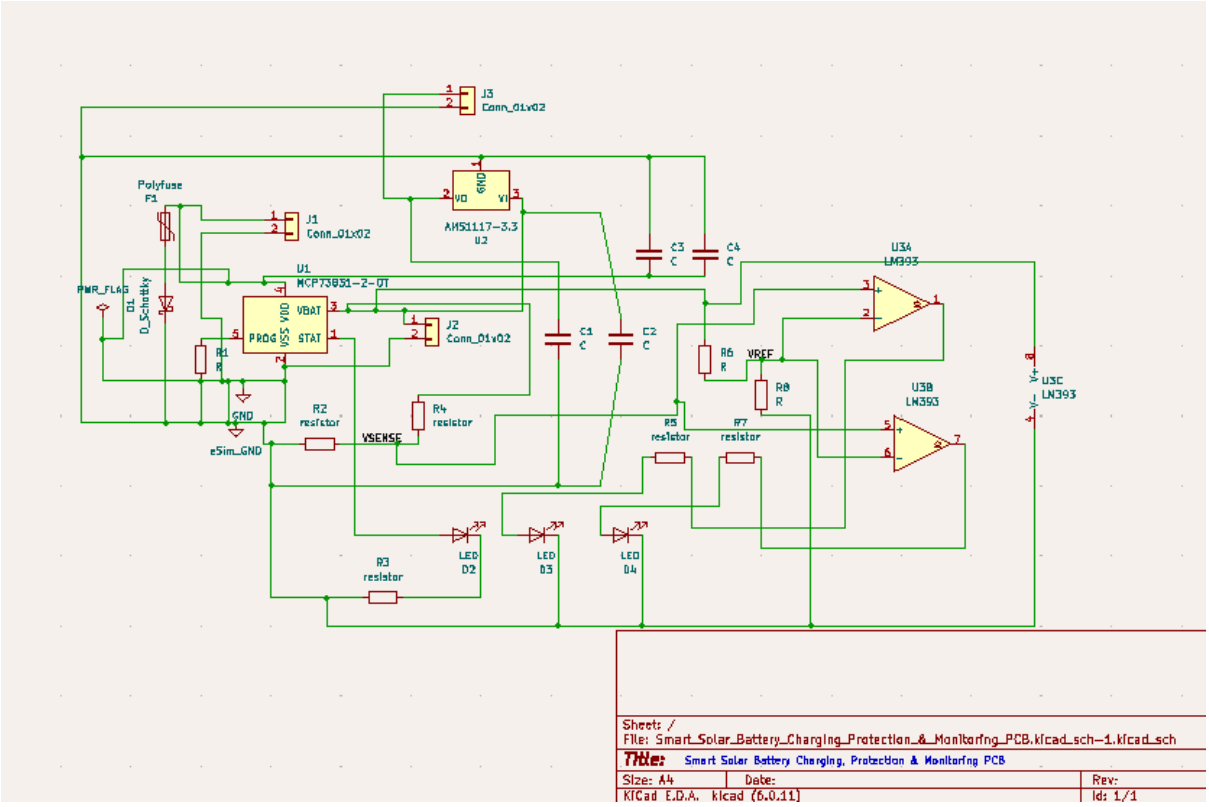
- To design a functional power-management-oriented PCB
- To create a clean and organized schematic using eSchema
- To assign suitable physical footprints to schematic components
- To transfer the schematic connectivity into the PCB editor
- To design the route a PCB using at least two copper layers
- To follow basic component placement and routing practices
- To perform DRC and resolve reported design issues
- To verify the final PCB using the 3D viewer in eSim

4. Circuit Description

Block	**Key Components**	**Function**
Solar input	J1 (2-pin connector)	Receives power from the external solar panel
Input protection	F1 (resettable polyfuse), D1(Schottky diode)	Over-current and reverse-polarity protection
Battery charging	U1 (MCP73831-2-OT)	Regulates single-cell Li-ion charging current and voltage
Battery Connector	J2 (2-pin connector)	Connects the battery pack to the board
Voltage monitoring	R1,R2 (Divier) → VSENSE net; R10/R11 - style reference divider →	Scales battery voltage down for comparison

	VREF net	
Status indication	U3 (dual LM393 comparator), D2/D3/D4(LDSs), R3-R8 (current-limit + bias resistors)	Drives charging / full / fault LEDs based on VSENSE vs VREF
Output Regulation	U2 (AMS1117-3.3),C1-C4 (decoupling)	Produces a clean, stable 3.3 V output rail
Output & Test points	J3 (output connector)	Delivers regulated power to an external load; test points for probing

5. Schematic

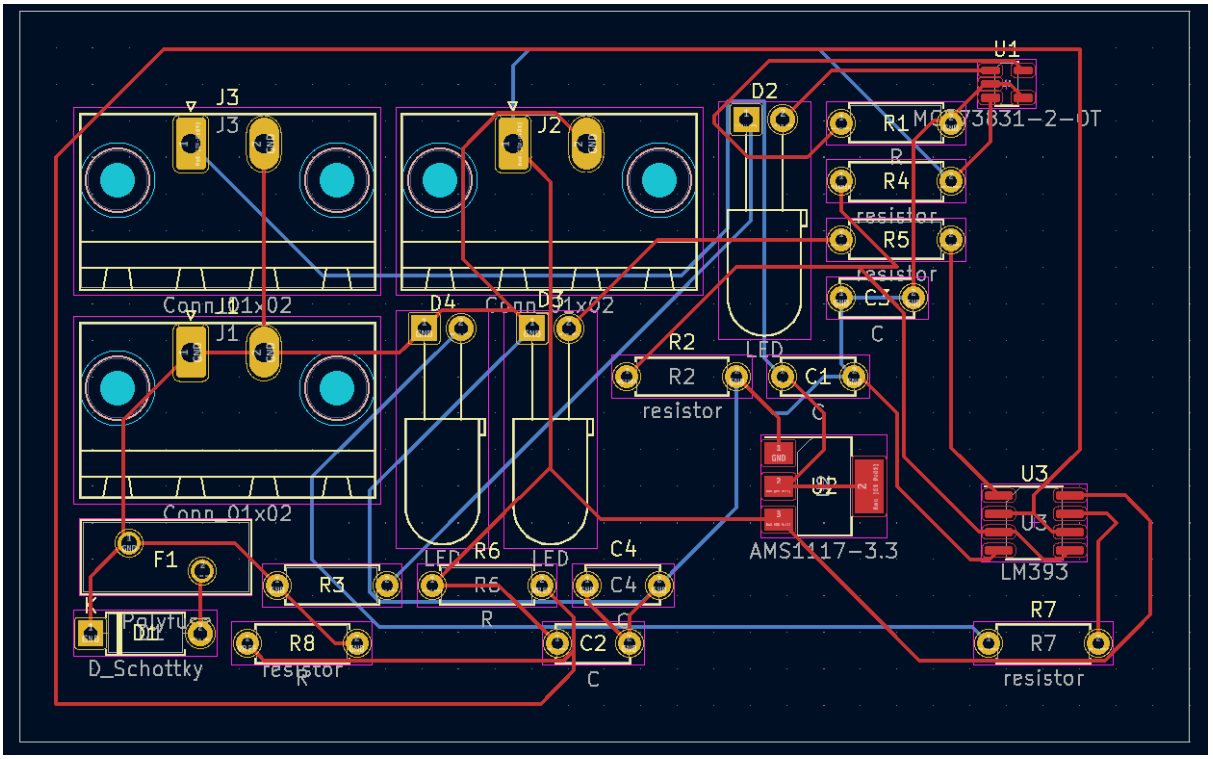


6. Footprint Assignment

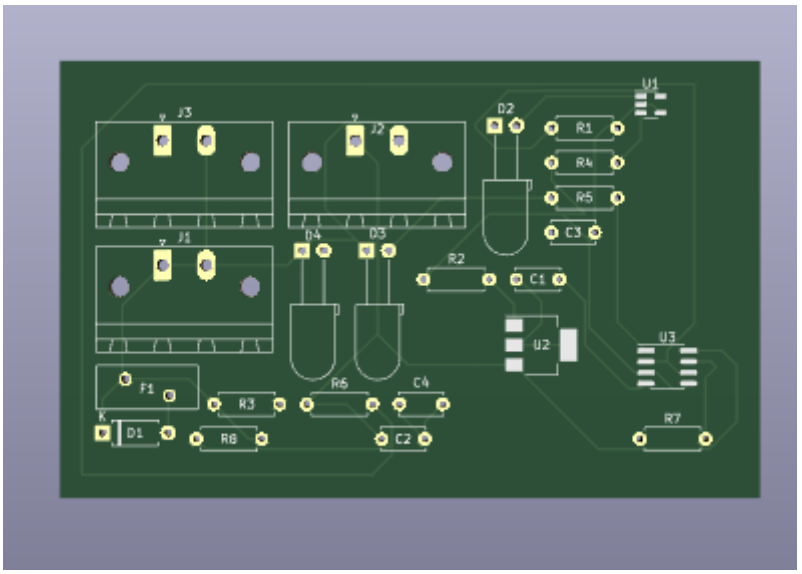
Component	Footprint assigned	Package type

C1-C4	Capacitor_THT:C_Disc_D5.0mm_W2.5mm_P5.00mm	THT ceramic disc
D1	Diode_THT:D_A-405_P7.62mm_Horizontal	THT axial (DO-41 size)
D2, D3, D4	LED_THT:LED_D5.0mm_Horizontal_O6.35mm_Z3.0mm	THT 5 mm LED
F1	Fuse:Fuse_BelFuse_0ZRE0075FF_L11.5mm_W4.8mm	THT resettable polyfuse
J1, J2, J3	Connector_Phoenix_MSTB:PhoenixContact_MSTBV_2,5_2-GF5,08_1x02_P5.00mm_Horizontal_ThreadedFlange_MountHole	THT screw terminal, 2-pin
R1–R8	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	THT axial 1/4 W
U1	Package_TO_SOT_SMD:SOT-23-5	SMD (from schematic library default)
U2	Package_TO_SOT_SMD:SOT-223-3_TabPin2	SMD (from schematic library default)
U3	Package_SO:SOIC-8_3.9x4.9mm_P1.27mm	SMD, 8-pin SOIC

7. PCB Layout



8. 3D View



9. DRC Results

Errors: 0

Warnings : 0